Week 10

Introduction

In the rapidly evolving landscape of electric vehicles (EVs) and renewable energy systems, the efficiency, reliability, and safety of power electronics are paramount. This week's lessons delve into the intricate world of designing critical magnetic components—transformers and inductors—which are the unsung heroes ensuring stable and efficient power conversion. Beyond component design, we explore the sophisticated communication protocols that govern EV charging, moving from the foundational principles of AC charging to the advanced high-power DC charging systems.

Understanding these topics is crucial for engineers and enthusiasts alike, as they form the backbone of modern charging infrastructure and power management. From optimizing the physical dimensions and performance of a transformer in an isolated DC-DC converter to ensuring a safe and intelligent dialogue between an EV and its charging station, this week's content highlights the complex interplay of electrical engineering, material science, and communication technology that underpins our electrified future.

Core Concepts

Term	Definition	Significance
`Peak AC Flux Density (deltaB)`	The maximum change in magnetic flux density within a transformer core during an AC cycle.	A critical design parameter that directly influences both core and copper losses, requiring careful optimization for efficiency.
`Air Gap (LG)`	A small physical gap intentionally introduced into the magnetic core of an inductor.	Crucial for preventing core saturation, increasing saturation current, and making inductance less dependent on core material permeability.
`Onboard Charger`	A power conversion unit located within the EV,	Converts incoming AC power from the grid into DC power suitable for the battery,

Term	Definition	Significance
	comprising AC-DC and isolated DC-DC converters.	ensuring isolation and proper voltage/current.
`CP Pin`	Control Pilot pin, a dedicated pin in the charging connector used for PWM-based communication.	Facilitates low-level communication between the EV and EVSE to manage the charging sequence and convey status.
`PP Pin`	Proximity Pilot pin, a dedicated pin in the charging connector used for detecting plug-in and cable current capacity.	Ensures a correct and safe connection by identifying the charger plug and its maximum current rating.
`PWM Communication`	Pulse Width Modulation-based communication, a method where information is encoded by varying the duty cycle or amplitude of a square wave.	Enables robust, low-level communication between the EV and EVSE, especially for AC charging, and serves as a carrier for PLC in DC charging.
`Power Line Communication (PLC)`	A technology that superimposes high-frequency data signals onto existing power lines (e.g., the Control Pilot line) for data transmission.	Facilitates high-level, bidirectional communication between the EV and EVSE, crucial for managing complex DC charging parameters.
`CCS2`	Combined Charging System Type 2, a universal connector standard that supports both AC and DC charging.	Allows for versatile charging options using a single vehicle inlet, simplifying infrastructure and user experience.
`Battery Management System (BMS)`	An electronic system within the EV that monitors and manages the battery pack's	Acts as the "master" during charging, communicating battery status and charging readiness to the EVSE.

Term	Definition	Significance
	voltage, current, and temperature.	

Detailed Analysis

1. Transformer Design for Isolated DC-DC Converters

The **Transformer Design for Isolated DC-DC Converter - III** lesson focuses on the iterative process of designing transformers to minimize losses and optimize physical dimensions. The core challenge lies in balancing two opposing loss mechanisms:
* **Core Loss (`P_core`)**: Energy dissipated in the magnetic core due to hysteresis and eddy currents, proportional to `deltaB` raised to the power `beta` . It increases with `deltaB` .
* **Copper Loss (`P_CU_total`)**: Energy dissipated in the windings due to wire resistance, inversely proportional to `deltaB` squared. It decreases as `deltaB` increases because fewer turns allow for thicker wires.

The design procedure aims to find an optimal `deltaB` where total losses (`P_total = P_core + P_CU_total`) are minimized. This involves:
* **Step 1: Core Selection**: Start with a candidate core (e.g., EE25) and obtain its physical (`Ac` , `WA` , `MLT` , `Lc`) and material (`beta` , `K_core`) properties from datasheets.
* **Step 2: Loss Evaluation**: Express `P_core` and `P_CU_total` as functions of `deltaB` , considering optimal winding area allocation (`gamma j`).
* **Step 3: `deltaB` Optimization**: Differentiate `P_total` with respect to `deltaB` and set to zero to find `deltaB_optimum` .
* **Step 4: Total Loss Check**: Calculate `P_total` at `deltaB_optimum` and compare it against `P_total_allowed` . If `P_total` exceeds the limit, a larger core must be selected, and the process repeated.
* **Step 5: Turns Calculation**: Once an optimal core and `deltaB` are found, calculate `N1 = lambda 1 / (2 * deltaB_optimum * Ac)` and other turns using specified ratios.
* **Step 6: Wire Size Determination**: Calculate required wire cross-sectional area (`AW_j`) based on `gamma j_optimum` , `Ku` , `WA` , and `N_j` , then select a standard wire gauge.
A final design check involves calculating magnetizing inductance, peak current, and winding resistances to verify total losses. Advanced considerations like AC effects (skin/proximity) and thermal management are crucial for high-frequency applications.

2. Filter Inductor Design

The **Filter Inductor Design** lesson focuses on designing inductors for smoothing current ripples in DC-DC converters. A key aspect is the role of the air gap:
* **Magnetic Circuit Fundamentals**: Without an air gap, inductance `L = μc * AC * N² / LC`. However, such cores are prone to saturation.
* **Role of Air Gap**: Introducing an air gap (`LG`) significantly increases the total reluctance, making the inductance primarily dependent on the air gap dimensions and number of turns (`L ≈ μ₀ * AG * N² / LG`), rather than the core material's permeability. This "stretches" the BH curve, increasing the saturation current and allowing the inductor to handle large DC bias currents without saturating.
* **Losses**: For filter inductors, the ripple current is small, making core losses and AC winding resistance effects often negligible compared to copper losses.
* **Design Outcomes**: The design aims to determine core size, number of turns (`N`), wire size (`AW`), and air-gap length (`LG`).
* **Constraints**: Four main equations govern the design:
1. Saturation Constraint: `N * I_max = B_max * LG / μ₀` (ensures `B_max < B_sat`).
2. Inductance Value: `L = μ₀ * AC * N² / LG` .
3. Winding Area Constraint: `ku * WA ≥ N * AW` .
4. Winding Resistance: `R = ρ * N * MLT / AW` .
* **Design Procedure**:
1. **Select Core**: Use the inequality `AC² * WA / MLT ≥ (ρ * L² * I_max²) / (B_max² * R * ku)` to select the smallest core that meets the requirements.
2. **Calculate N**: `N = (L * I_max) / (B_max * AC)` .
3. **Determine LG**: `LG = (μ₀ * AC * N²) / L` .
4. **Evaluate AW**: `AW ≤ (ku * WA) / N` , then select a standard wire gauge.
A final design check verifies that copper losses (`I_rms² * R`) are within the `P_allow` limit. If not, a larger core is needed.

3. AC Type-2 Charging - I: Architecture and Initial Communication

The **AC Type-2 Charging - I** lesson introduces the architecture and initial communication protocols for AC charging.
* **System Architecture**:
* **EVSE (ACVSE)**: Provides AC power from the grid, contains a controller, sensors, and safety interlocks (relays/contactors).
* **EV Internal Components**: Includes the **Onboard Charger** (AC-DC and isolated DC-DC converters) for power conversion, and the **BMS** (Battery Management System) which acts as the "master" for charging control.
* **Communication Protocols**:
* **Low-Level PWM Communication**: Occurs over the **CP Pin** (Control Pilot) using a

1 kHz Pulse Width Modulation signal. The EVSE generates the PWM, and the EV interprets its duty cycle for maximum current. The EV signals readiness by changing the PWM voltage level.
* **Proximity Pilot (PP) Pin**: Crucial for initial detection and safety. It identifies the charger plug and its maximum current capacity.
* **AC Type-2 and CCS2 Connectors**:
* **AC Type-2**: Standard connector with L1, L2, L3, N, PE, CP, and PP pins.
* **CCS2 (Combined Charging System Type 2)**: A versatile standard where the top portion is compatible with AC Type-2 plugs for AC charging, and the full connector (including DC+ and DC- pins) is used for DC charging.
* **PP Circuit and Cable Detection**: The **PP Pin** circuit uses a specific resistance (R6) within the charger plug to signal the cable's current capacity to the EV. The EV measures the voltage at the PP pin (VPP) to determine the R6 value and thus the cable's rating. For example, R6 = 1500 Ω corresponds to 13A AC charging (VPP ≈ 3.87V), while R6 = 100 Ω corresponds to 63A AC charging (VPP ≈ 1.14V). This prevents the EV from drawing more current than the cable can safely handle.

4. AC Type-2 Charging - II: Detailed Communication States

Building on the previous lesson, **AC Type-2 Charging - II** elaborates on the detailed communication states and current control mechanisms of AC charging.
* **CP Circuit Communication**: The EVSE generates a 1 kHz PWM signal (+12V to -12V). The EV measures its duty ratio and frequency. The EV signals its readiness by closing an internal S2 switch, which connects a resistor (R2) in parallel with R3, changing the voltage level of the positive PWM pulse.
* **Charging Sequence States**:
* **State A (Not Connected)**: CP at +12V.
* **State B1 (EV Connected, Not Ready)**: CP at +9V (PWM off). EVSE detects connection.
* **State B2 (EV Connected, EVSE Ready)**: CP at +9V (PWM on). EV detects PWM, knows EVSE is ready.
* **State C2 (EV and EVSE Ready - Charging)**: EV closes S2 (R2 in parallel with R3), CP at +6V (PWM on). EVSE detects +6V, AC power transfer begins.
* **State D (EV Ready, Requires Ventilation)**: EV closes S2 with a different resistor, CP at +3V (PWM on).
* **State E (Fault Condition)**: CP at 0V.
* **State F (EVSE Unavailable)**: CP at -12V.
* **Duty Ratio and Current Control**: The duty ratio of the PWM signal directly communicates the maximum current the EVSE can provide.
* `3% - 7%`: Reserved for DC charging (high-level communication).
* `10% - 85%`: Max current = 0.6 * Duty Ratio.

* `85% - 96%`: Max current = (Duty Ratio - 64) * 2.5.
The EV's **BMS** ensures that the current drawn never exceeds this indicated maximum. The EVSE can dynamically adjust the duty ratio.

5. DC Charging (CCS2) - I: High-Level Communication

The **DC Charging (CCS2)-I** lesson introduces the complexities of DC fast charging, which requires high-level communication due to the external power conversion unit in the EVSE.
* **Contrast with AC Charging**: In DC charging, the EVSE directly supplies DC power to the battery. This necessitates extensive data exchange between the EV's **BMS** and the EVSE's controller (SECC) regarding battery conditions (SOC, voltage, current, temperature).
* **CCS2 Connector for DC**: The **CCS2** connector integrates both AC Type 2 and DC pins. When a full CCS2 plug is inserted, the **CP Pin** still carries a 1 kHz PWM signal, but its duty ratio is specifically maintained between 3% and 7%. This signals to the EV that it is a DC CCS2 charger, prompting the activation of high-level communication.
* **Power Line Communication (PLC)**: High-level communication is achieved by superimposing high-frequency data signals onto the 1 kHz PWM signal on the **CP Pin**. Both EV and EVSE have PLC modems.
* **Standard**: Follows the HomePlug Green PHY standard, a subset of HomePlug AV.
* **SLAC (Signal Level Attenuation Characterization)**: Performed initially to ensure a stable logical connection between PLC modems.
* **Modulation**: Uses OFDM (Orthogonal Frequency Division Multiplexing) over a 2-28 MHz frequency band, making it robust against interference.
* **Data Rate**: 4.5 to 10 Mbps.
* **Signal Amplitude**: Small (0.3Vpp to 1.3Vpp) superimposed on the PWM.
* **DC CCS2 Charging Sequence (Initial Stages)**:
* **T0: Mating and B1 State**: Plug inserted, EV detects via **PP Pin**, immobilizes, enters B1 state (+9V, PWM off).
* **T0-T1: Initialization (B2 State)**: EVSE activates PWM (3-7% duty ratio), signaling DC charging. PLC handshaking begins. EV sends max voltage/current limits. Safety check for DC voltage > 60V.
* **T1-T2: Connector Lock and Compatibility Check**: EV locks connector. EVSE responds with capabilities, performs insulation check.
* **T2-T3: Cable Check Stage (C2/D State)**: EV changes CP state to C2 (or D), requests cable/insulation check. EVSE continuously checks isolation.
* **T3-T4: Cable Check Completion**: EVSE confirms isolation > 100 kΩ, indicates "Valid" and "Ready."

* **T5: Pre-charge Stage**: Contactors close. EV (master) sends "Pre-Charge Request" with desired DC current (initially < 2A) and target DC voltage.

Key Takeaways

- [ ] Transformer design involves an iterative process to balance core and copper losses by optimizing `deltaB`.
- [ ] The optimal `deltaB` minimizes total transformer losses, where core loss increases with `deltaB` and copper loss decreases with `deltaB`.
- [ ] Air gaps are crucial in filter inductors to prevent core saturation and make inductance largely independent of core material permeability.
- [ ] Filter inductor design prioritizes copper loss minimization, as core losses are often negligible due to small ripple currents.
- [ ] AC charging relies on the EV's onboard charger for power conversion, with the EV's BMS acting as the master controller.
- [ ] The **PP Pin** circuit uses resistance (R6) to detect the connected cable's current capacity, ensuring safety in AC charging.
- [ ] The **CP Pin** uses a 1 kHz PWM signal for low-level communication, with voltage levels indicating charging states and duty ratio signaling maximum current.
- [ ] DC charging requires high-level **Power Line Communication (PLC)** over the **CP Pin** due to external power conversion and extensive data exchange with the EV's **BMS**.
- [ ] The **CCS2** connector provides a unified interface for both AC and DC charging, with a specific PWM duty ratio (3-7%) signaling DC mode.
- [ ] DC charging involves a multi-stage sequence with critical safety checks, including SLAC, connector locking, cable checks, and pre-charging.

Conclusion

This week's exploration into transformer and inductor design, coupled with the intricate world of EV charging protocols, underscores the sophisticated engineering required to power our modern world. From the meticulous balancing of magnetic and electrical losses in a transformer to the precise control of current in a filter inductor, the design of these fundamental components directly impacts the efficiency and reliability of power conversion systems. These principles are then extended to the dynamic realm of electric vehicle charging, where safety and intelligent communication are paramount.

The transition from basic AC charging, with its low-level PWM communication, to the advanced high-level Power Line Communication of DC fast charging, illustrates a

continuous drive towards greater efficiency, speed, and safety. Understanding the roles of the **PP Pin**, **CP Pin**, and the **BMS**, along with the multi-stage charging sequences, provides a comprehensive view of how EVs and charging stations engage in a complex, yet seamless, dialogue. As the demand for faster and more reliable charging infrastructure grows, the mastery of these design and communication principles will be essential for engineers shaping the future of sustainable transportation and energy management.